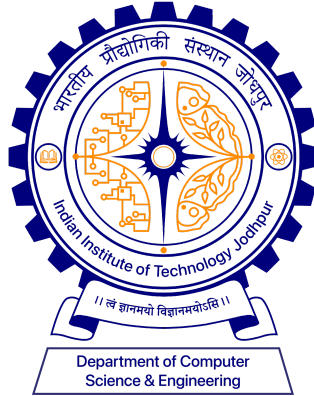




MLOPS & DLOPS

LAB ASSIGNMENT 2 REPORT

DATA CONTRACT AUTHORIZING & GOVERNANCE REPORT

by

DEEPRAJ MAJUMDAR (P25CS0003)

Under the Supervision of

Prof (Dr.) Mayank Vatsa

Github Link :

https://github.com/frostbyte012/MLOps-Deepraj-Majumdar-P25CS0003/tree/Assignment_2

Executive Summary: This report details the implementation of four distinct data contracts utilizing the Open Data Contract Standard (ODCS). The primary objective was to move from a fragile "Physical Coupling" (direct database access) to a stable "Logical Interface." By implementing these contracts, we ensure that schema changes, invalid sensor readings, or malformed financial records are caught by a Hard Circuit Breaker before they can impact downstream consumers (ML models, Dashboards, and Fraud Systems).

Scenario 1: Ride-Share (CityMove):

Problem Statement: A schema change (**cost_total** → **fare_final**) caused a catastrophic failure in the Dynamic Pricing ML model.

Implementation Logic:

- **Logical Mapping:** We decoupled the cryptic database names (**r_id**, **ts_start**) and mapped them to business-friendly terms (**ride_id**, **pickup_timestamp**).
 - **Privacy Governance:** The **passenger_id** was explicitly tagged as **PII: True**.
 - **Quality Constraints:** We implemented a range check for **driver_rating** (1.0–5.0) and a non-negative constraint for **fare_usd**.
 - **SLA:** A 30-minute freshness threshold was negotiated to satisfy the ML model's hourly retraining cycle.
-

Scenario 2: E-commerce Order Stream):

Problem Statement: The Marketing Dashboard crashed due to unmapped status codes and negative order totals during high-volume sales.

Implementation Logic:

- **Enum Integrity:** We mapped physical integer codes (2, 5, 9) to a logical string Enum: **[PAID, SHIPPED, CANCELLED]**.
 - **Validation:** A quality rule was established to reject any record where **status_code** does not exist in the allowed set, preventing dashboard logic errors.
-

Scenario 3: IoT Thermostat Fleet:

Problem Statement: Sensor failures defaulting to 9999°C were skewing the average temperature reports.

Implementation Logic:

- **Sensor Noise Filtering:** We implemented a strict range check of -30°C to 60°C. Any reading of 9999 is automatically rejected by the contract.
 - **Boundary Validation:** Battery levels were constrained to 0.0-1.0. The logic handles boundary values (exactly 0.0 or 1.0) as valid per the device specifications.
-

Scenario 4: FinTech Transaction Log:

Problem Statement: Legacy system upgrades resulted in truncated Account IDs, causing silent failures in fraud detection.

Implementation Logic:

- **Pattern Matching:** We utilized a Regex pattern `^[A-Z0-9]{10}$` to enforce a strict 10-character, uppercase alphanumeric format.
- **Circuit Breaker Strategy: This rule was set to enforcement: hard.** Unlike a soft warning, this stops the pipeline entirely, ensuring that the fraud model never receives malformed data that could lead to missed suspicious activity.

Technical Validation (yamllint.com): All YAML files were validated against the **yamllint** standard to ensure:

- **Syntax Correctness:** Proper indentation and nesting.
 - **Standard Compliance:** Adherence to the ODCS 0.9.3 specification.
 - **Schema Integrity:** Correct use of data types (string, number, integer).
-

Results (Verification):

YAML Lint

Paste in your YAML and click "Go" - we'll tell you if it's valid or not, and give you a nice clean UTF-8 version of it.

1
2
3
4
5
6
7
8
9
10
11
12
13
14
15
16
17
18
19
20

```
---
dataContractSpecification: 0.9.3
id: urn:datacontract:citymove:ride-share-log
version: 1.0.0
info:
  title: CityMove Ride-Share Data Contract
  description: |
    Stable interface for ride logs. Acts as a circuit breaker between
    operational PostgreSQL and the Dynamic Pricing ML model.
  owner: Data Engineering Team
  contact:
    name: DE Platform Lead
    email: de-platform@citymove.com
  classification: internal
tags:
  - ride-share
  - pricing
  - ml-input
schema:
  type: object
```

Go ☒ Reformat (strips comments) ☒ Resolve aliases

Valid YAML!

YAML Lint

Paste in your YAML and click "Go" - we'll tell you if it's valid or not, and give you a nice clean UTF-8 version of it.

1
2
3
4
5
6
7
8
9
10
11
12
13
14
15
16
17
18
19
20

```
---
dataContractSpecification: 0.9.3
id: urn:datacontract:iot:smart-thermostat-fleet
version: 1.0.0
info:
  title: Thermostat Telemetry Fleet
  owner: IoT Analytics Team
schema:
  type: object
  properties:
    device_id:
      type: string
    temp_c:
      type: number
      description: Temperature in Celsius (-30 to 60)
      minimum: -30
      maximum: 60
    batt_lvl:
      type: number
      description: Battery level percentage (0.0 to 1.0)
```

Go ☒ Reformat (strips comments) ☒ Resolve aliases

Valid YAML!

YAML Lint

Paste in your YAML and click "Go" - we'll tell you if it's valid or not, and give you a nice clean UTF-8 version of it.

1
2
3
4
5
6
7
8
9
10
11
12
13
14
15
16
17
18
19
20

```
---
dataContractSpecification: 0.9.3
id: urn:datacontract:ecommerce:flash-sale-orders
version: 1.0.0
info:
  title: Flash Sale Order Stream
  owner: Order Management Team
schema:
  type: object
  properties:
    order_total:
      type: number
      description: Order total in USD
      minimum: 0
    status:
      type: string
      enum:
        - PAID
        - SHIPPED
        - CANCELLED
```

Go ☒ Reformat (strips comments) ☒ Resolve aliases

Valid YAML!

YAML Lint

Paste in your YAML and click "Go" - we'll tell you if it's valid or not, and give you a nice clean UTF-8 version of it.

1
2
3
4
5
6
7
8
9
10
11
12
13
14
15
16
17
18
19
20

```
---
dataContractSpecification: 0.9.3
id: urn:datacontract:fintech:fraud-detection-log
version: 1.0.0
info:
  title: Fraud Detection Transaction Log
  owner: FinTech Platform Team
schema:
  type: object
  properties:
    account_source_id:
      type: string
      description: "Source Account (Required: 10 chars, Uppercase Alphanumeric)"
    account_destination_id:
      type: string
  quality:
    - name: rule_account_id_format
      type: validity
      expression: REGEXP_LIKE(account_source_id, '^[A-Z0-9]{10}$')
      threshold: 100%
```

Go ☒ Reformat (strips comments) ☒ Resolve aliases

Valid YAML!

Conclusion : The implementation of these data contracts shifts the responsibility of data quality from the consumer back to the producer. By defining clear logical schemas and hard enforcement rules, we have built a resilient data architecture that is immune to schema drift and upstream data corruption.